

Name: _____ Partner: _____ Period: _____

Missing Data Points

A scientist's measurement device records four data streams, each modeled by a rational function. At certain input values, the device has a gap in the data. Your job is to analyze each function, determine whether each gap is a **hole** or a **vertical asymptote**, and — if it is a hole — find the exact coordinates.

The four functions are:

$$f_1(x) = \frac{x^2 - 9}{x - 3} \quad f_2(x) = \frac{x^2 - 1}{(x - 1)(x + 4)} \quad f_3(x) = \frac{(x - 2)^2}{x^2 - 4} \quad f_4(x) = \frac{x^2 + 2x}{x}$$

Tier R — Remediate (Key)

Fn	Numerator (factored)	Denominator (factored)	Gap at	Hole or VA?
f_1	$(x - 3)(x + 3)$	$(x - 3)$	$x = 3$	Hole
f_2	$(x - 1)(x + 1)$	$(x - 1)(x + 4)$	$x = 1$	Hole
			$x = -4$	VA
f_3	$(x - 2)^2$	$(x - 2)(x + 2)$	$x = 2$	Hole
			$x = -2$	VA
f_4	$x(x + 2)$	x	$x = 0$	Hole

Tier A — Apply (Key)

1. Factored forms and multiplicity analysis:

- $f_1: \frac{(x-3)(x+3)}{(x-3)}$ — shared zero $x = 3$, $m = n = 1$, so **hole**.
- $f_2: \frac{(x-1)(x+1)}{(x-1)(x+4)}$ — shared zero $x = 1$, $m = n = 1$, so **hole**; $x = -4$ in denom only, so **VA**.
- $f_3: \frac{(x-2)^2}{(x-2)(x+2)}$ — shared zero $x = 2$, $m = 2 > n = 1$, so **hole**; $x = -2$ in denom only, so **VA**.
- $f_4: \frac{x(x+2)}{x}$ — shared zero $x = 0$, $m = n = 1$, so **hole**.

2. Hole coordinates:

Function	Shared zero c	Simplified form	Hole at (c, L)
f_1	3	$x + 3$	(3, 6)
f_2	1	$\frac{x+1}{x+4}$	$\left(1, \frac{2}{5}\right)$
f_3	2	$\frac{x-2}{x+2}$	(2, 0)
f_4	0	$x + 2$	(0, 2)

Tier E — Extend (Key)

1. Limit notation:

$$\lim_{x \rightarrow 3} f_1(x) = 6 \quad \lim_{x \rightarrow 1} f_2(x) = \frac{2}{5} \quad \lim_{x \rightarrow 2} f_3(x) = 0 \quad \lim_{x \rightarrow 0} f_4(x) = 2$$

2. Sample explanation for f_1 : **The hole is at (3, 6) because after canceling the shared factor $(x - 3)$ from numerator and denominator, the simplified form is $x + 3$. Substituting $x = 3$ gives $3 + 3 = 6$, which is the output the function approaches.**

3. Sample construction: $g(x) = \frac{(x-1)(x+2)}{(x-1)(x+2)^2} \cdot k$ for some nonzero k , or more simply:

Factored: $g(x) = \frac{(x-1)}{(x-1)(x+2)}$

Simplified: $g(x) = \frac{1}{x+2}$ with hole at $\left(1, \frac{1}{3}\right)$

(Any rational function with $(x-1)$ in both numerator and denominator, and $(x+2)$ only in the denominator after cancellation, is correct.)